CONVOLUTION

Topic 4

Recap Image Enhancement

- Image enhancement: Improving image features such as edges, boundaries, contrast, and brightness
- Image enhancement approaches: spatial domain and frequency domain.
- Spatial domain methods operate directly on image pixels, i.e., deal with the manipulation of pixel values.
- Methods of pixel manipulation include: Point, Mask, Global Operation
- Point operation include • Brightness • Contrast • Histogram
- Point operation methods include: • Contrast stretching • Histogram • Thresholding
- • Image negatives • Intensity level stretching • Bit-plane slicing

Image Enhancement (cont')

- **Spatial Filtering**

  - The use of spatial masks for image processing usually is called *spatial filtering*, and the masks themselves are called *spatial filters*.
  - In this section we consider linear and nonlinear spatial filters for image enhancement.
  - So-called *lowpass* filters attenuate or eliminate high-frequency components in the Fourier domain while leaving low frequencies untouched (that is, the filter "passes" low frequencies).
  - High-frequency components characterize edges and other sharp details in an image, so the net effect of *lowpass* filtering is image blurring.
  - Similarly, *highpass* filters attenuate or eliminate low-frequency components.

Image Enhancement (cont')

- **Spatial Filtering**
  - Because these components are responsible for the slowly varying characteristics of an image, such as overall contrast and average intensity, the net result of *highpass* filtering is a reduction of these features and a correspondingly apparent sharpening of edges and other sharp details.
  - A third type of filtering, called *bandpass filtering*, removes selected frequency regions between low and high frequencies.

Filters Summary

Filter Type	Definition	Frequency Range Affected	Effect on Image	Common Use Cases
Lowpass	Allows low-frequency components to pass; attenuates high-frequency ones.	Low frequencies	Smooths image, reduces sharp transitions and noise.	Noise reduction, smoothing, anti-aliasing
Highpass	Allows high-frequency components to pass; suppresses low-frequency ones.	High frequencies	Enhances edges and fine details.	Edge detection, feature extraction
Bandpass	Allows mid-frequency components to pass; blocks low and high frequencies.	Mid-range frequencies	Highlights textures, removes blur and noise.	Texture analysis, pattern recognition

Image Enhancement (cont')

- **Spatial Filtering**
  - The figure below shows cross sections of circularly symmetric lowpass, highpass, and bandpass filters in the frequency domain and their corresponding spatial filters.
  - The horizontal axes for the figures in the top row correspond to frequency, and their counterparts in the bottom row are spatial coordinates.
  - The shapes in the bottom row are used as guidelines for specifying linear spatial *I* filters.

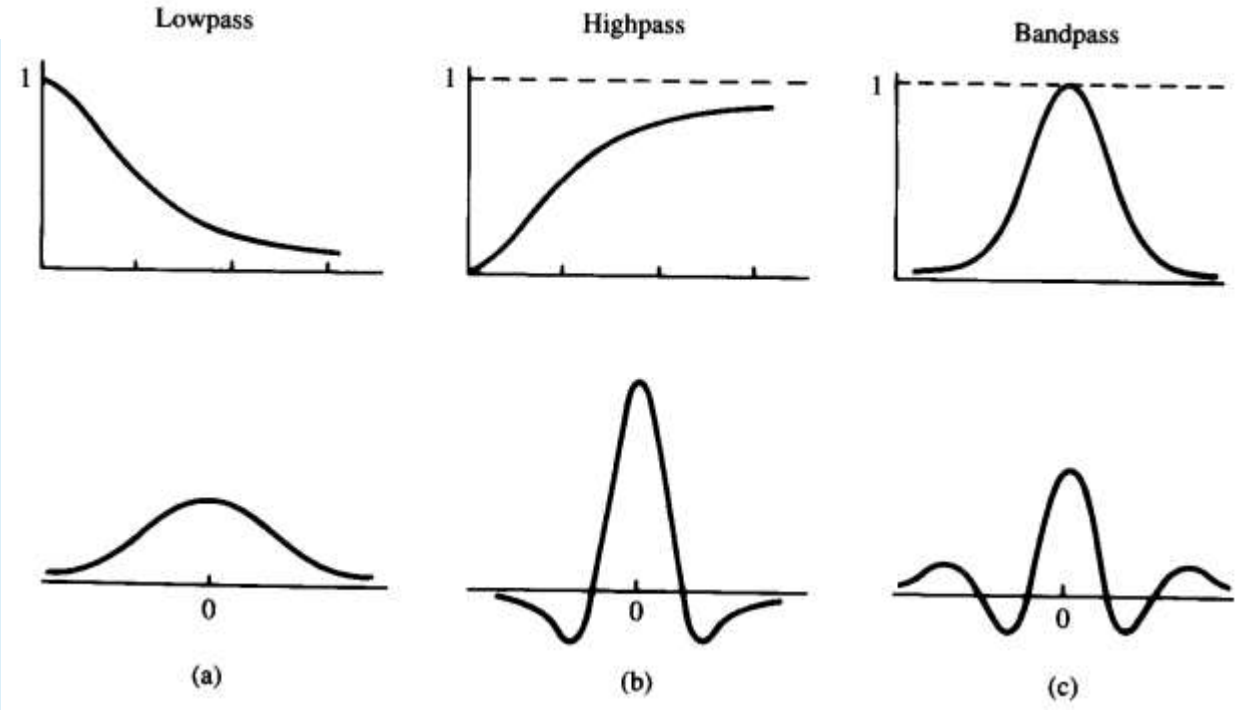


Image Enhancement (cont')

- **Spatial Filtering**
  - Regardless of the type of linear filter used, however, the basic approach is to sum products between the mask coefficients and the intensities of the pixels under the mask at a specific location in the image.
  - The figure below shows a general 3 × 3 mask.
  - Denoting the gray levels of pixels under the mask at any location by $z_1, z_2, ..., z_9$, the response of a linear mask is

$$R = w_1z_1 + w_2z_2 + ... + w_9z_9$$

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

A 3 × 3 mask with arbitrary coefficients (weights)

Image Enhancement (cont')

- **Spatial Filtering**
  - If the center of the mask is at location (x, y) in the image, the gray level of the pixel located at (x, y) is replaced by R.
  - The mask is then moved to the next pixel location in the image and the process is repeated.
  - This continues until all pixel locations have been covered. The value of R is computed by using partial neighborhoods for pixels that are located in the border of the image.
  - Also, usual practice is to create a new image to store the values of R, instead of changing pixel values in place.
  - This practice avoids using gray levels that have been altered as a result of an earlier application of this equation.

- **Spatial Filtering**

Example:

Original Image

0	0	0	0	0
0	0	0	0	0
0	0	1	0	0
0	0	0	0	0
0	0	0	0	0

Mask

1	1	1
1	5	1
1	1	1

center

Map the center of the mask to the pixel of the original image

1	1	1
1	5	1
1	1	1

0	0	0	0	0
0	0	0	0	0
0	0	1	0	0
0	0	0	0	0
0	0	0	0	0

New Image

0	0	0	0	0
0	1	1	1	0
0	1	5	1	0
0	1	1	1	0
0	0	0	0	0

Result of filtered image

Image Enhancement (cont')

- **Spatial Filtering**
  - Nonlinear spatial filters also operate on neighborhoods.
  - Not using sum operation
  - In general, however, their operation is based directly on the values of the pixels in the neighborhood under consideration, and they do not explicitly use mask weightings.
  - Noise reduction can be achieved effectively with a nonlinear filter whose basic function is to compute the median gray-level value in the neighborhood in which the filter is located.
  - Other examples include the max filter (with a response $R = \max\{z_k \mid k = 1, 2, ..., 9\}$), which is used to find the brightest points in an image, and the min filter, which is used for the opposite purpose.
  - As examples to the spatial filtering technique, we will briefly cover some methods of the *smoothing filter* and the *sharpening filter*.

Image Enhancement (cont')

- **Smoothing Filter**

  - Smoothing filters are used for blurring and for noise reduction.
  - Blurring is used in preprocessing steps, such as removal of small details from an image prior to (large) object extraction, and bridging of small gaps in lines or curves.
  - Noise reduction can be accomplished by blurring with a linear filter and also by nonlinear filtering.
  - We will cover two types of smoothing filter:

    1) Spatial Lowpass Filter (linear)
    2) Median Filter (non-linear)

Image Enhancement (cont')

- **Spatial Lowpass Filter**
  - The shape of the impulse response needed to implement a lowpass (smoothing) spatial filter indicates that the filter has to have all positive coefficients.
  - For a 3×3 spatial filter, the simplest arrangement would be a mask in which all coefficients have a value of 1 (Figure (a) next page).
  - However, the response would then be the sum of gray levels for nine pixels, which could cause R to be out of the valid gray-level range.
  - The solution is to scale the sum by dividing R by 9. Larger masks follow the same concept, as Figures (b) and (c) show.
  - Note that, in all these cases, the response R would simply be the average of all the pixels in the area of the mask.
  - For this reason, the use of masks shown next page is often referred to as *neighborhood averaging*.

Image Enhancement (cont')

- **Spatial Lowpass Filter**

Spatial lowpass filters of various sizes.

$$\frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

(a)

$$\frac{1}{25} \times \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

(b)

$$\frac{1}{49} \times \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

(c)

- **Spatial Lowpass Filter**

(a) Original image; (b)—(f) results of spatial lowpass filtering with a mask of size $n \times n$, $n = 3$, 5, 7, 15, 25.

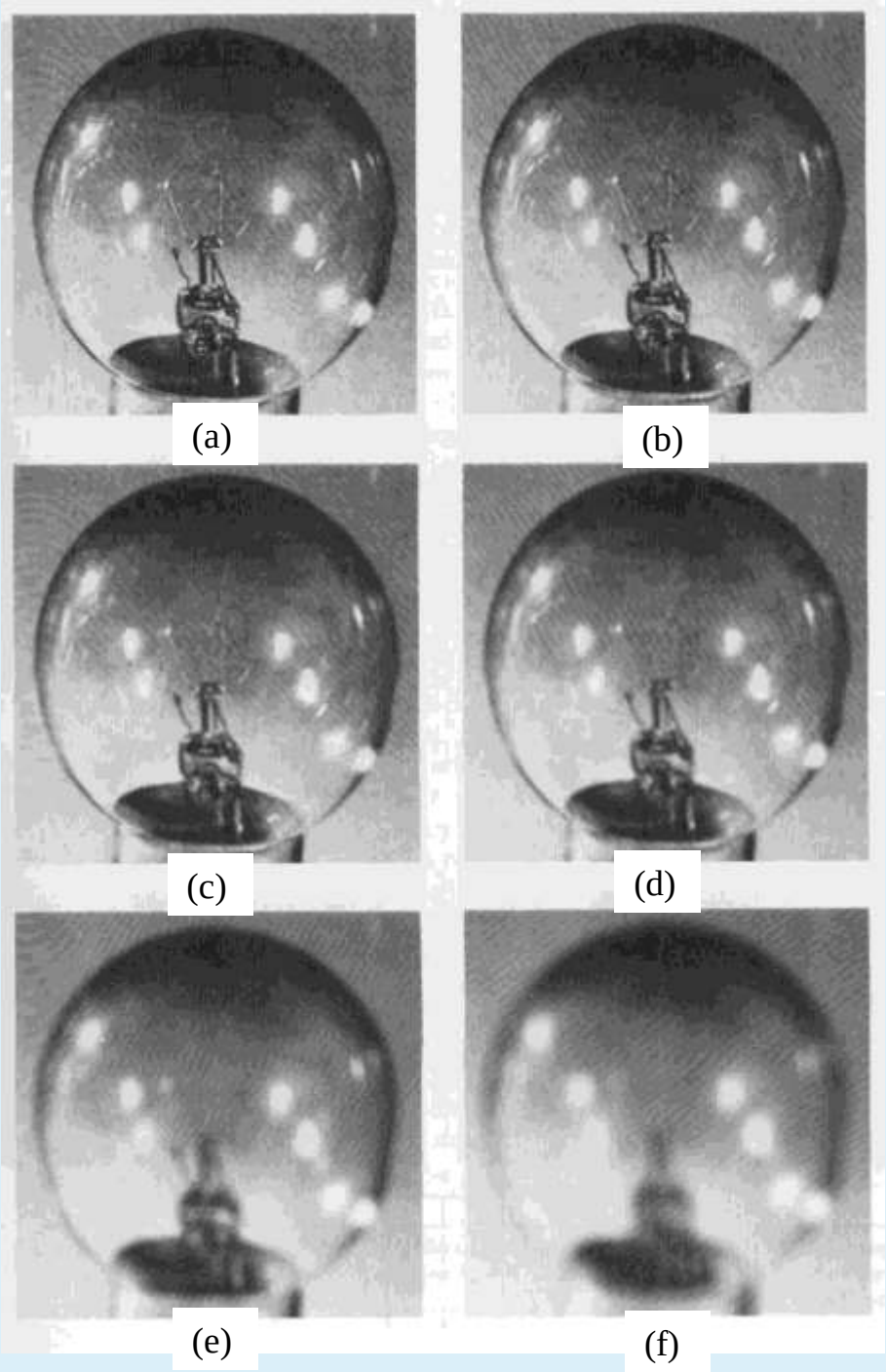


Image Enhancement (cont')

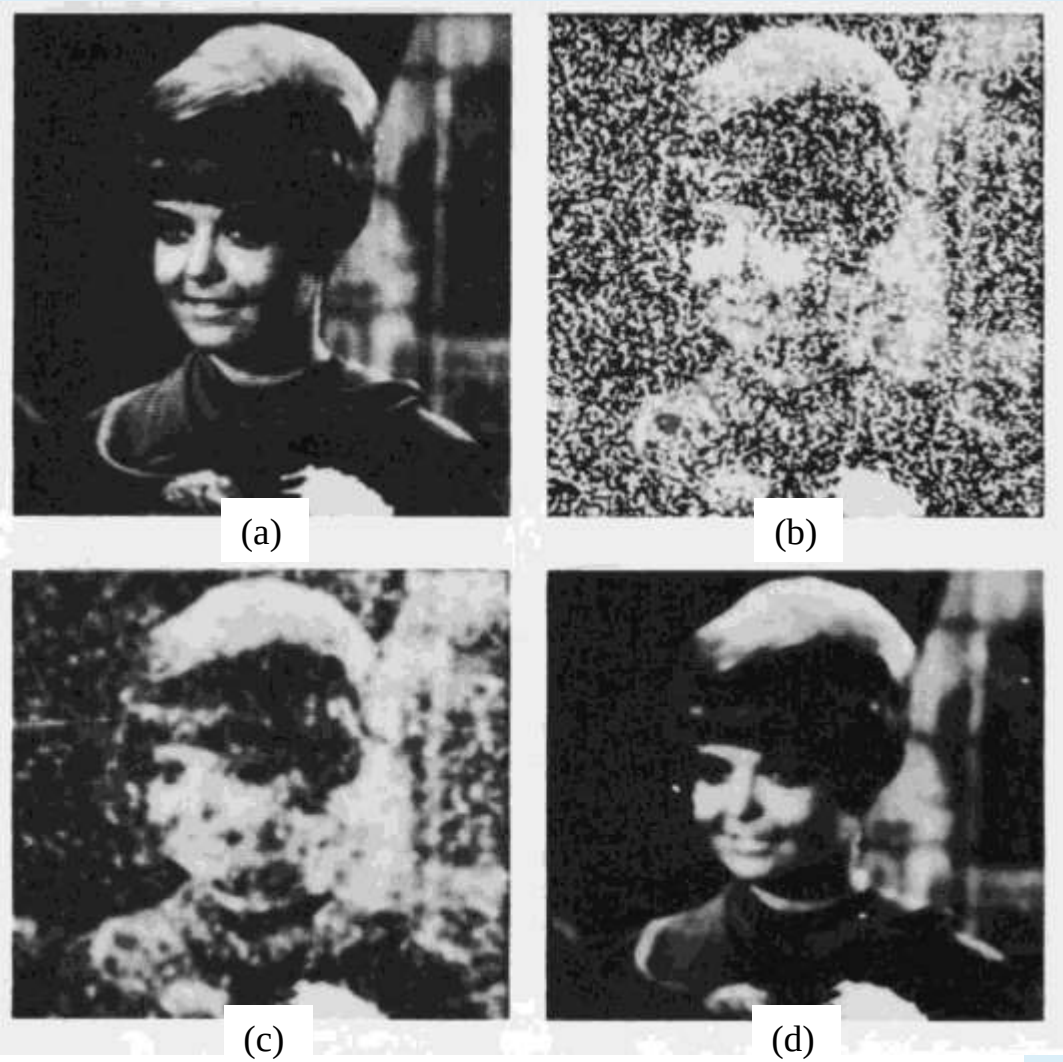
- **Median Filter**
  - One of the principal difficulties of the smoothing method discussed in the preceding section is that it blurs edges and other sharp details.
  - If the objective is to achieve noise reduction rather than blurring, an alternative approach is to use *median filters*.
  - That is, the gray level of each pixel is replaced by the median of the gray levels in a neighborhood of that pixel, instead of by the average.
  - This method is particularly effective when the noise pattern consists of strong, spike like components and the characteristic to be preserved is edge sharpness.
  - *Median filters* are nonlinear.

Image Enhancement (cont')

- **Median Filter**

- The median *m* of a set of values is such that half the values in the set are less than *m* and half are greater than *m*.
- First sort the values of the pixel and its neighbors, determine the median, and assign this value to the pixel.
- Example, in a 3×3 neighborhood the median is the 5th largest value, in a 5×5 neighborhood the 13th largest value, and so on.
- When several values in a neighborhood are the same, all equal values have to be grouped. For example, suppose that a 3×3 neighborhood has values (10, 20, 20, 20, 15, 20, 20, 25, 100). These values are sorted as (10, 15, 20, 20, 20, 20, 20, 25, 100), which results in a median of 20.
- Thus, the principal function of median filtering is to force points with distinct intensities to be more like their neighbors, actually eliminating intensity spikes that appear isolated in the area of the filter mask.
- Eliminates salt-and-pepper noise; preserves edges; but erodes corners

- **Median Filter**



(a) Original image; (b) image corrupted by impulse noise; (c) result of 5×5 neighborhood averaging, (d) result of 5×5 median filtering.

- **Median Filter**

Example:

123	125	126	130	140
122	124	126	127	135
118	120	150	125	134
119	115	119	123	133
111	116	110	120	130

Neighbourhood values:

115, 119, 120, 123, 124,
125, 126, 127, 150

Median value: 124

- **Median Filter**

Example:

123	125	126	130	140
122	124	126	127	135
118	120	150	125	134
119	115	119	123	133
111	116	110	120	130

Neighbourhood values:

118, 120, 122, 123, 124, 126, 126, 126, 150

Median value = 124

- **Median Filter**

Example:

123	125	126	130	140
122	124	126	127	135
118	120	150	125	134
119	115	119	123	133
111	116	110	120	130

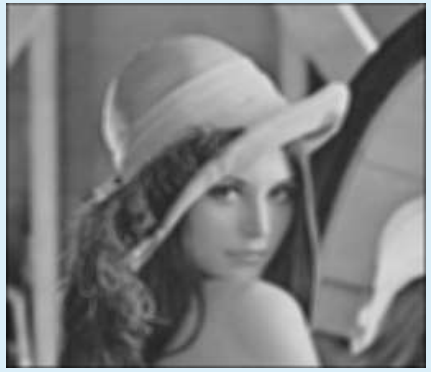
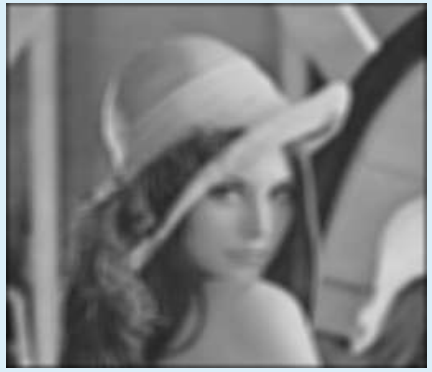
Neighbourhood values:

110, 111, 115, 116, 118, 119, 119, 120, 150

Median value = 118

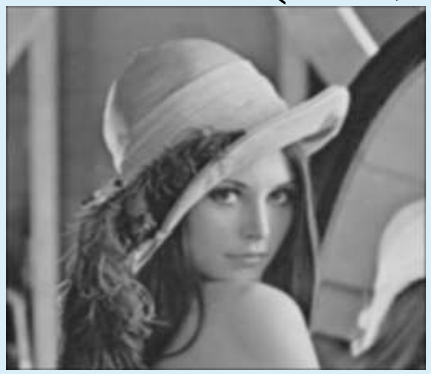
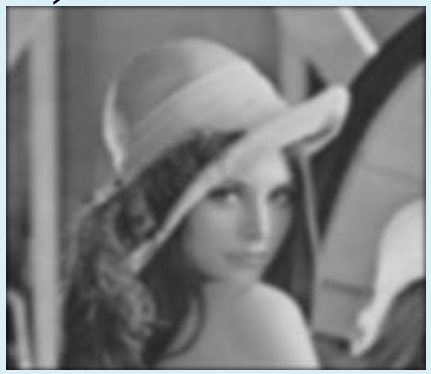
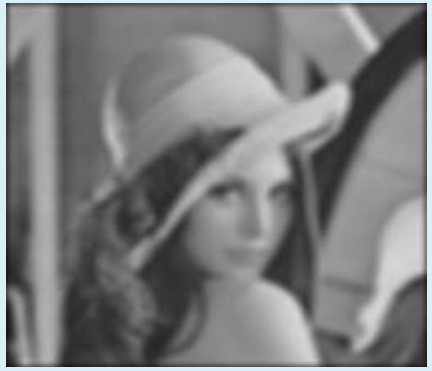
Examples:

Spatial Lowpass (3, 5, 7):

Input image:

Gaussian (σ=1, 2, 3):

Median (3, 5, 7):

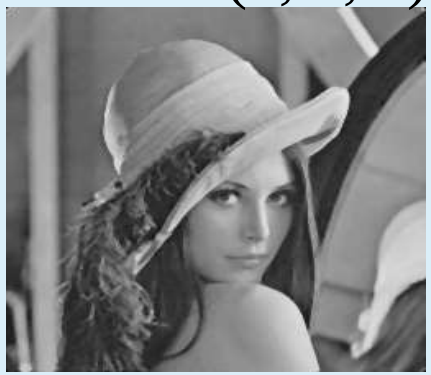


Image Enhancement (cont')

- **Sharpening Spatial Filters**

  - The principal objective of sharpening is to highlight fine details in an image or to enhance detail that has been blurred, either in error or as a natural effect of a particular method of image acquisition.
  - Uses of image sharpening vary and include applications ranging from electronic printing and medical imaging to industrial inspection and autonomous target detection in smart weapons.
  - We will briefly look into these 3 examples of sharpening filters:

    1) Spatial Highpass Filter
    2) High-boost Filtering
    3) Derivative Filtering

Image Enhancement (cont')

- **Spatial Highpass Filter**

- The shape of the impulse response needed to implement a highpass (sharpening) spatial filter indicates that the filter should have positive coefficients near its center, and negative coefficients in the outer periphery.
- For a 3 × 3 mask, choosing a positive value in the center location with negative coefficients in the rest of the mask meets this condition.
- The figure below shows an implementation of a 3 × 3 sharpening filter. Note that the sum of the coefficients is 0. Thus, when the mask is over an area of constant or slowly varying gray level, the output of the mask is zero or very small.

$$\frac{1}{9} \times \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Image Enhancement (cont’)

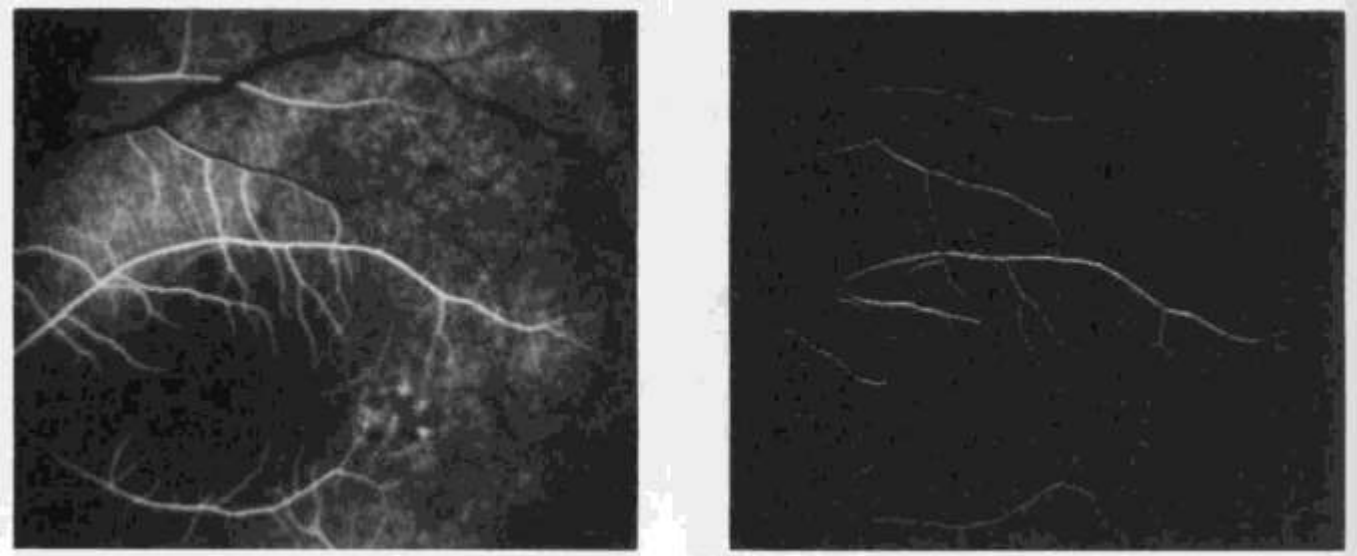
- **Spatial Highpass Filter**

  - Note also that this filter eliminates the zero-frequency term.
  - Figure (b) shows the result of using the highpass filter on the image of Figure (a).
  - The original image consists of fairly fine detail over a significant portion of slowly varying background levels.
  - The result is characterized by somewhat enhanced edges over a rather dark background.
  - Significantly better results can be obtained by using high-boost filtering, which is discussed in the next section.

- **Spatial Highpass Filter**

$$\frac{1}{9} \times \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

A basic highpass spatial filter.



(a) (b)

(a) Image of a human retina; (b) highpass filtered result using the mask shown on top of page.

Image Enhancement (cont')

- **High-boost Filter**

  - A highpass filtered image may be computed as the difference between the original image and a lowpass filtered version of that image; that

$$Highpass = Original - Lowpass\ .$$

  - Multiplying the original image by an amplification factor, denoted by A, yields the definition of a *high-boost* or *high-frequency-emphasis filter*:

$$\begin{aligned} High\text{-}boost &= (A)(Original) - Lowpass, \\ &= (A)(Original) - Original + Original - Lowpass, \\ &= (A - 1)(Original) + Original - Lowpass \\ &= (A - 1)(Original) + Highpass. \end{aligned}$$

  - An $A = 1$ value yields the standard highpass result. When $A > 1$, part of the original is added back to the highpass result, which restores partially the lowfrequency components lost in the highpass filtering operation.

Image Enhancement (cont')

- **High-boost Filter**

  - The result is that the high-boost image looks more like the original image, with a relative degree of edge enhancement that depends on the value of *A*.

  - In terms of implementation, the preceding results can be combined by letting the center weight of the mask to be:

$$w = 9A - 1$$

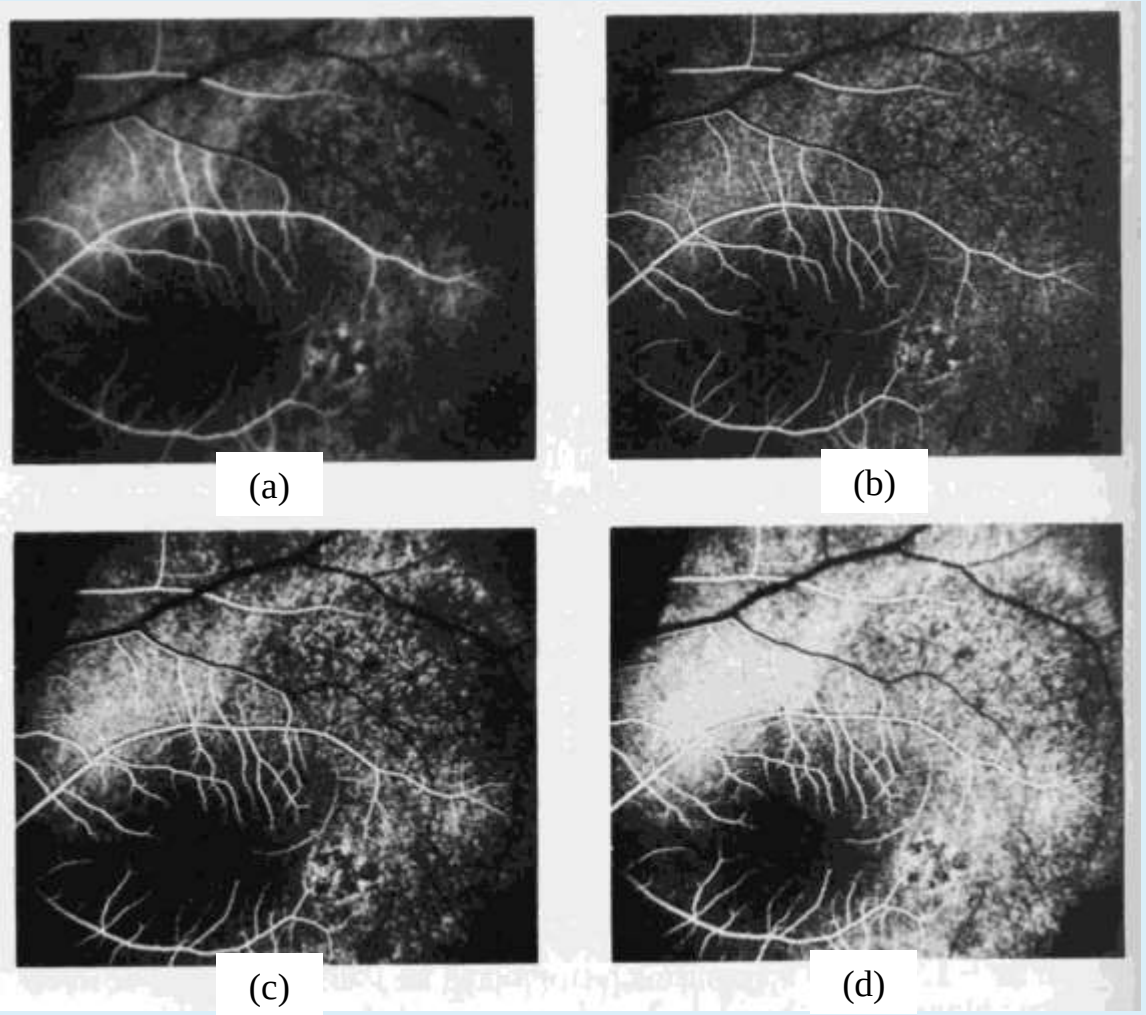
    with $A \geq 1$.

  - The value of *A* determines the nature of the filter.

- **High-boost Filter**

$$\frac{1}{9} \times \begin{bmatrix} -1 & -1 & -1 \\ -1 & w & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Mask used for high-boost spatial filtering. The value of the center weight is $w = 9A - 1$, with $A \geq 1$.



(a) Original image; (b)—(d) result of high-boost filtering using the mask shown above, with A = 1.1, 1.15, and 1.2, respectively. Compare these results with those produced by the spatial highpass filter.

Image Enhancement (cont')

- **Derivative Filter**

  - Averaging of pixels over a region tends to blur detail in an image.
  - As averaging is analogous to integration, differentiation can be expected to have the opposite effect and thus sharpen an image.
  - The most common method of differentiation in image processing applications is the gradient.
  - For a function $f(x, y)$, the gradient of f at coordinates (x, y) is defined as the vector

$$\nabla f \equiv grad(f) \equiv \begin{bmatrix} g_x \\ g_y \end{bmatrix} \equiv \begin{bmatrix} \dfrac{\partial f}{\partial x} \\ \dfrac{\partial f}{\partial y} \end{bmatrix}$$

Image Enhancement (cont')

- **Derivative Filter**

  - The magnitude (length) of this vector,

$$M(x,y) = mag(\nabla f) = \sqrt{g_x^2 + g_y^2}$$

  is the basis for various approaches to image differentiation.

  - In some implementation, it is more suitable computationally to approximate the squares and square root operations by absolute value:

$$M(x,y) \approx |g_x| + |g_y|$$

Image Enhancement (cont')

- **Derivative Filter**

  - Consider the image region shown in the figure below, where the z's denote the intensities of image points in 3 × 3 region.

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

A 3 × 3 region of an image (the z's are intensity values)

  - Approximations to g_x and g_y using a 3 × 3 neighborhood centered on z_5 are as follows:

Image Enhancement (cont')

- **Derivative Filter**

$$g_x = \frac{\partial f}{\partial x} = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)$$

and

$$g_y = \frac{\partial f}{\partial x} = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)$$

- After computing the derivative in *x* and *y* direction, we obtain the magnitude of the gradient:

$$M(x, y) = \left|(z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)\right| + \left|(z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)\right|$$

- The masks in figure (next slide) are called the *Sobel operators*.

- **Derivative Filter**

-1	*-2*	*-1*
0	*0*	*0*
1	*2*	*1*

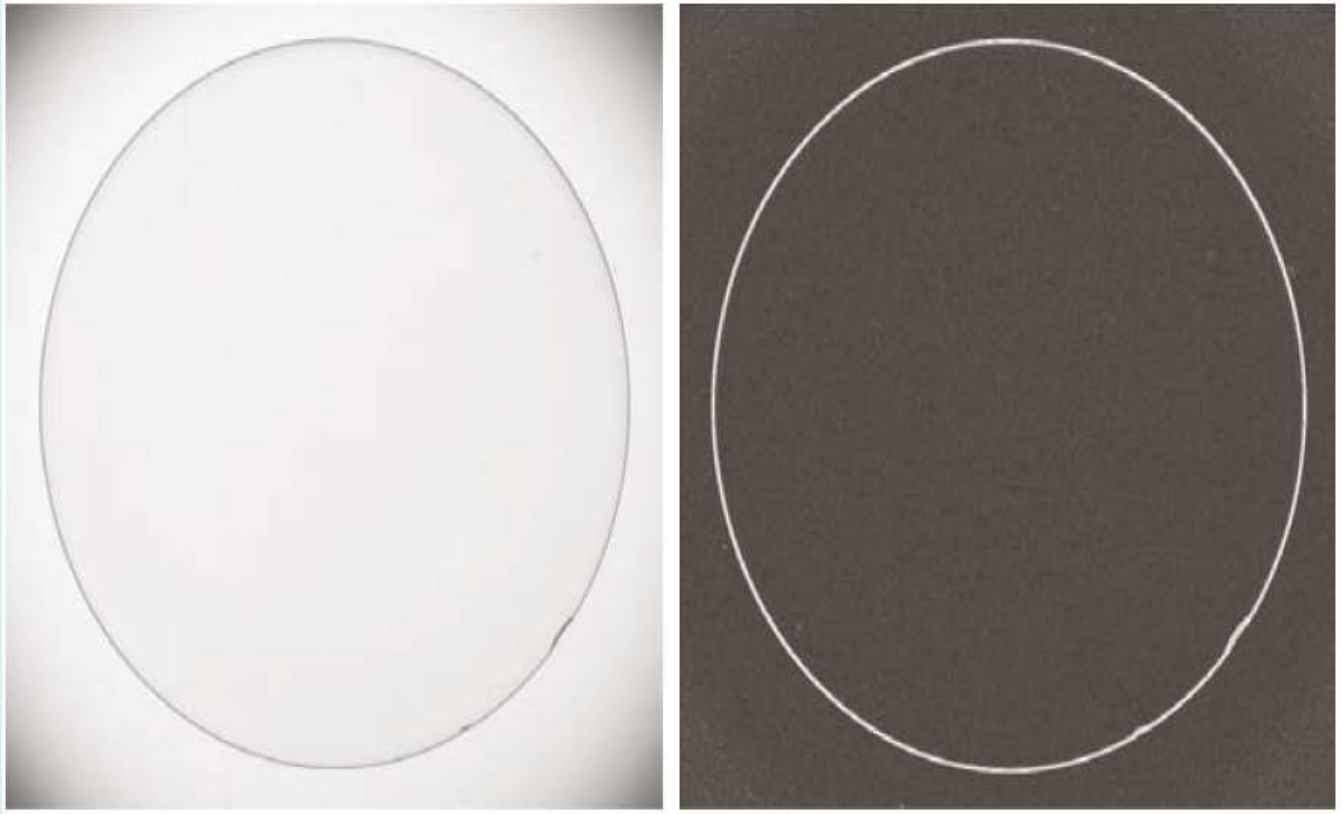
x-direction

-1	*0*	*1*
-2	*0*	*2*
-1	*0*	*1*

y-direction

Commonly used first-order derivative filter: Sobel operators

Example:



(a) (b)

(a) Optical image contact lens (note defects on the boundary at 4 and 5 o'clock). (b) *Sobel* gradient.